

Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 4: Roots of Polynomials

Chapter 4: Roots of Polynomials

Why this matters

This chapter reveals a beautiful shortcut: you can find relationships between the roots of a polynomial (their sum, sum of products in pairs, product, and more) directly from its coefficients — without ever solving the equation. Combined with the ability to transform an equation into a new one with related roots, this is one of the most elegant and exam-friendly topics in the whole course.

Concept: sum and products of roots

For a quadratic $ax^2 + bx + c = 0$ with roots α and β , you already know $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$. Further Maths extends this pattern to cubics and quartics. For a cubic $ax^3 + bx^2 + cx + d = 0$ with roots α, β, γ :

These results are not arbitrary — they fall straight out of expanding the factorised form. Since the cubic has roots α, β, γ , it can be written $a(x - \alpha)(x - \beta)(x - \gamma) = ax^3 + bx^2 + cx + d$. Expanding the brackets gives $a[x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \alpha\gamma)x - \alpha\beta\gamma]$. Comparing coefficients of x^2 , x and the constant term with $bx^2 + cx + d$ directly produces $\Sigma\alpha = -b/a$, $\Sigma\alpha\beta = c/a$ and $\alpha\beta\gamma = -d/a$ — the same comparing-coefficients idea used throughout this course.

$$\begin{aligned}\Sigma\alpha &= \alpha + \beta + \gamma = -b/a \\ \Sigma\alpha\beta &= \alpha\beta + \beta\gamma + \alpha\gamma = c/a \\ \alpha\beta\gamma &= -d/a\end{aligned}$$

For a quartic $ax^4 + bx^3 + cx^2 + dx + e = 0$ with roots $\alpha, \beta, \gamma, \delta$, the pattern of alternating signs continues:

$$\Sigma\alpha = -b/a, \Sigma\alpha\beta = c/a, \Sigma\alpha\beta\gamma = -d/a, \alpha\beta\gamma\delta = e/a$$

Key vocabulary

Term	Meaning
$\Sigma\alpha$	Sum of all the roots
$\Sigma\alpha\beta$	Sum of all products of pairs of roots
$\Sigma\alpha\beta\gamma$	Sum of all products of triples of roots (quartics only)
$\alpha\beta\gamma$ (or $\alpha\beta\gamma\delta$)	Product of all the roots
Symmetric function of the roots	An expression in the roots unchanged by swapping any two roots, e.g. $\alpha^2 + \beta^2 + \gamma^2$

Transformed equation	A new polynomial whose roots are a specified function of the original roots
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Worked example 1: finding root relationships from a cubic
EXAMPLE 1

Question. The cubic $2x^3 - 5x^2 + 3x - 7 = 0$ has roots α, β, γ . Write down the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \alpha\gamma$, and $\alpha\beta\gamma$.

Step 1. Identify $a = 2, b = -5, c = 3, d = -7$. **(B1)**

Step 2. $\Sigma\alpha = -b/a = -(-5)/2 = 5/2$. **(A1)**

Step 3. $\Sigma\alpha\beta = c/a = 3/2$. **(A1)**

Step 4. $\alpha\beta\gamma = -d/a = -(-7)/2 = 7/2$. **(A1)**

Answer: $\alpha + \beta + \gamma = 5/2, \alpha\beta + \beta\gamma + \alpha\gamma = 3/2, \alpha\beta\gamma = 7/2$.

Exam tip
EXAM TIP

Say the pattern out loud as "alternate the signs, and divide by a each time, starting with $-b/a$ ". Writing out the general cubic formulae from memory (rather than looking them up) every single time builds the fluency needed for the harder symmetric-function questions later.

Concept: symmetric functions of the roots

A huge family of exam questions asks for expressions like $\alpha^2 + \beta^2 + \gamma^2$, or $1/\alpha + 1/\beta + 1/\gamma$, in terms of $\Sigma\alpha, \Sigma\alpha\beta$ and $\alpha\beta\gamma$. The key trick is almost always to expand $(\Sigma\alpha)^2$ and rearrange, since $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \alpha\gamma)$.

Worked example 2: a symmetric function
EXAMPLE 2

Question. Using the roots of $2x^3 - 5x^2 + 3x - 7 = 0$ from Worked example 1, find $\alpha^2 + \beta^2 + \gamma^2$.

Step 1. Start from the identity $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \alpha\gamma)$, so $\alpha^2 + \beta^2 + \gamma^2 = (\Sigma\alpha)^2 - 2\Sigma\alpha\beta$. **(M1)**

Step 2. Substitute $\Sigma\alpha = 5/2$ and $\Sigma\alpha\beta = 3/2$: $\alpha^2 + \beta^2 + \gamma^2 = (5/2)^2 - 2(3/2) = 25/4 - 3$. **(M1 A1)**

Step 3. Simplify: $25/4 - 12/4 = 13/4$. **(A1)**

Answer: $\alpha^2 + \beta^2 + \gamma^2 = 13/4$.

Another pattern worth memorising: since $1/\alpha + 1/\beta + 1/\gamma = (\alpha\beta + \beta\gamma + \alpha\gamma)/(\alpha\beta\gamma) = \Sigma\alpha\beta/\alpha\beta\gamma$, any symmetric expression involving reciprocals of the roots can be written directly in terms of $\Sigma\alpha, \Sigma\alpha\beta$ and $\alpha\beta\gamma$ without ever finding the individual roots.

Concept: transforming equations

Given a polynomial with roots α , β , γ , we can find a new polynomial whose roots are a related quantity (e.g. 2α , 2β , 2γ , or $1/\alpha$, $1/\beta$, $1/\gamma$, or $\alpha + 1$ etc.) using a substitution:

1. Let $y = f(x)$ be the new root in terms of the old one.
2. Rearrange to make x the subject in terms of y .
3. Substitute this expression for x into the original equation.
4. Clear any fractions and rearrange into standard descending-power form.

Worked example 3: transforming an equation (roots + k)

EXAMPLE 3

Question. The equation $x^3 - 4x^2 + x - 6 = 0$ has roots α , β , γ . Find the equation with roots $\alpha + 2$, $\beta + 2$, $\gamma + 2$.

Step 1. Let $y = x + 2$, so $x = y - 2$. (M1)

Step 2. Substitute into the original equation: $(y-2)^3 - 4(y-2)^2 + (y-2) - 6 = 0$. (M1)

Step 3. Expand each term: $(y-2)^3 = y^3 - 6y^2 + 12y - 8$; $4(y-2)^2 = 4(y^2 - 4y + 4) = 4y^2 - 16y + 16$. (M1 A1)

Step 4. Combine everything: $y^3 - 6y^2 + 12y - 8 - 4y^2 + 16y - 16 + y - 2 - 6 = 0$, giving $y^3 - 10y^2 + 29y - 32 = 0$. (M1 A1)

Answer: The equation with roots $\alpha + 2$, $\beta + 2$, $\gamma + 2$ is $y^3 - 10y^2 + 29y - 32 = 0$.

Exam tip

EXAM TIP

For a "roots + k" or "roots $\times$ k" transformation, you can (and should) check your answer using $\Sigma\alpha$ without doing the full substitution: here $\Sigma(\alpha+2)$ should equal $\Sigma\alpha + 6 = 4 + 6 = 10$, which matches $-(-10)/1 = 10$ from the new equation. This cross-check habit catches algebra slips before they cost the whole question.

Worked example 4: transforming an equation (reciprocal roots)

EXAMPLE 4

Question. The equation $x^3 + 3x^2 - 2x + 5 = 0$ has roots α , β , γ . Find the equation with roots $1/\alpha$, $1/\beta$, $1/\gamma$.

Step 1. Let $y = 1/x$, so $x = 1/y$. (M1)

Step 2. Substitute into the original equation: $(1/y)^3 + 3(1/y)^2 - 2(1/y) + 5 = 0$. (M1)

Step 3. Multiply every term by y^3 to clear denominators: $1 + 3y - 2y^2 + 5y^3 = 0$. (M1 A1)

Step 4. Rearrange into standard form (highest power first): $5y^3 - 2y^2 + 3y + 1 = 0$. (A1)

Answer: The equation with roots $1/\alpha$, $1/\beta$, $1/\gamma$ is $5y^3 - 2y^2 + 3y + 1 = 0$.

Exam tip

EXAM TIP

The "reciprocal roots" transformation is really just reversing the order of the coefficients (up to sign considerations) — spotting this pattern from the original 1, 3, -2, 5 becoming 5, -2, 3, 1 gives you a fast way to check your multiplication-through-by- y^n step.

Chapter summary

Idea	Key point
Cubic root relations	$\Sigma\alpha = -b/a$, $\Sigma\alpha\beta = c/a$, $\alpha\beta\gamma = -d/a$
Quartic root relations	Signs continue to alternate: $-b/a$, c/a , $-d/a$, e/a
Symmetric functions	Expand $(\Sigma\alpha)^2$ or similar identities to rewrite in terms of $\Sigma\alpha$, $\Sigma\alpha\beta$, $\alpha\beta\gamma$
Transforming equations	Substitute $x = f^1(y)$ into the original equation and clear fractions/expand
Checking transformations	Verify $\Sigma(\text{new roots})$ against the new equation's coefficients as a sanity check

Common exam mistakes

- Mixing up the sign pattern (forgetting it alternates: $-$, $+$, $-$, $+$...).
- Forgetting to divide by a when the leading coefficient is not 1.
- Expansion errors when substituting into transformed equations, especially with negative signs.
- Forgetting to multiply through by y^n (or the appropriate power) to clear all fractions in a reciprocal-roots transformation.
- Leaving the final transformed equation not in standard descending-power form.

You should now be able to

- State and use the relationships between the roots and coefficients of cubic and quartic equations.
- Find symmetric functions of the roots (such as $\Sigma\alpha^2$ or $\Sigma 1/\alpha$) using algebraic identities.
- Find a new polynomial equation whose roots are a given linear or reciprocal function of the original roots.
- Check a transformed equation using the root-coefficient relationships.